

Preface

The purpose of this book is to provide a tutorial overview on the subject of computerized tomographic imaging. We expect the book to be useful for practicing engineers and scientists for gaining an understanding of what can and cannot be done with tomographic imaging. Toward this end, we have tried to strike a balance among purely algorithmic issues, topics dealing with how to generate data for reconstruction in different domains, and artifacts inherent to different data collection strategies.

Our hope is that the style of presentation used will also make the book useful for a beginning graduate course on the subject. The desired prerequisites for taking such a course will depend upon the aims of the instructor. If the instructor wishes to teach the course primarily at a theoretical level, with not much emphasis on computer implementations of the reconstruction algorithms, the book is mostly self-contained for graduate students in engineering, the sciences, and mathematics. On the other hand, if the instructor wishes to impart proficiency in the implementations, it would be desirable for the students to have had some prior experience with writing computer programs for digital signal or image processing. The introductory material we have included in Chapter 2 should help the reader review the relevant practical details in digital signal and image processing. There are no homework problems in the book, the reason being that in our own lecturing on the subject, we have tended to emphasize the implementation aspects and, therefore, the homework has consisted of writing computer programs for reconstruction algorithms.

The lists of references by no means constitute a complete bibliography on the subject. Basically, we have included those references that we have found useful in our own research over the years. Whenever possible, we have referenced books and review articles to provide the reader with entry points for more exhaustive literature citations. Except in isolated cases, we have not made any attempts to establish historical priorities. No value judgments should be implied by our including or excluding a particular work.

Many of our friends and colleagues deserve much credit for helping bring this book to fruition. This book draws heavily from research done at Purdue by our past and present colleagues and collaborators: Carl Crawford, Mani Azimi, David Nahamoo, Anders Andersen, S. X. Pan, Kris Dines, and Barry Roberts. A number of people, Carl Crawford, Rich Kulawiec, Gary S. Peterson, and the anonymous reviewers, helped us proofread the manuscript;

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